

Trainings ▾

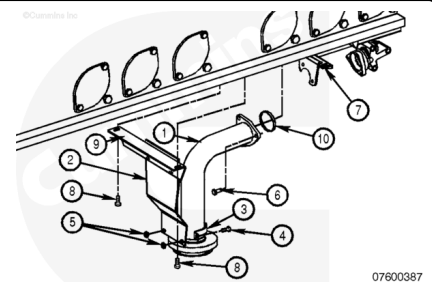
Remove

There are different lubricating oil suction tube arrangements depending on the oil pan style.

Remove the nuts (5), capscrews (4), and strap (3) from the vertical support bracket assembly (2).

Remove the capscrews (8) and remove the vertical support bracket assembly.

Remove the three capscrews (6), and the oil suction tube (1).



Clean and Inspect for Reuse

⚠ WARNING ⚠

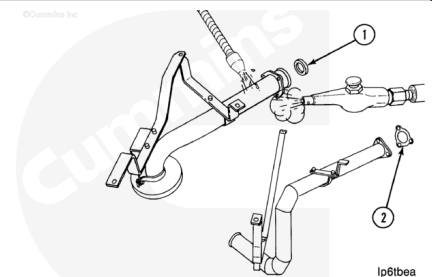
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Fly debris and dirt can cause personal injury.



The oil suction tube on the engine being serviced can be different than illustrated. Different oil sump locations and depths require different oil suction tubes. The type of screen

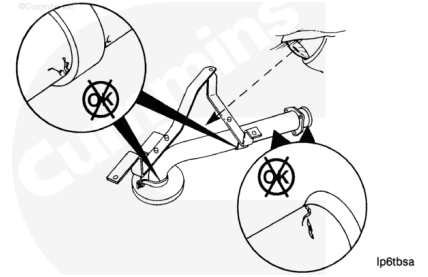
on the end of the lubricating oil suction tube can be different.

Remove the sealing ring (1), or the gasket (2) from the lubricating oil suction tube.

Use solvent or steam to clean the lubricating oil suction tube and dry with compressed air.

⚠ CAUTION ⚠

If the lubricating oil suction tube is repaired by welding, be sure all of the welding slag and any metal that can break are removed from the inside of tube. If debris enters the lubricating oil pump and system, the gears will be damaged and failure will result.



Check the lubricating oil suction tube and brackets for damage or cracks. Check for wear in the area of the support brackets.

If the lubricating oil suction tube or the brackets are cracked or damaged, they **must** be repaired or replaced.

Install

Install the small support bracket, if applicable (11).

Torque Value: 45 n•m [33 ft-lb]

Install the lubricating oil suction tube (1) and seal (10) to oil pump with the three capscrews (6).

Torque Value: 45 n•m [33 ft-lb]

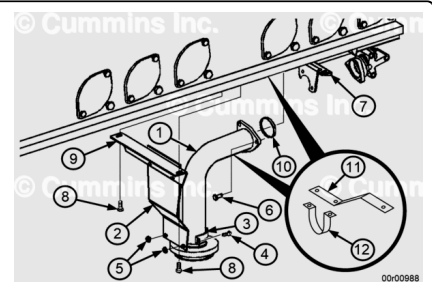
Note : If the rear housing has been removed and reinstalled, it will possibly be necessary to realign it when installing the lubricating oil suction tube.

Hand tighten the strap on the small bracket, if applicable (12).

Install the lubricating oil suction tube support bracket (2) and hand tighten the two to four supporting capscrews (8).

Note : The bracket may have two or four capscrews depending on the oil pan style.

In the slotted holes, the bolts **must** be at the tube end of the slot.



Install the vertical support bracket straps (3) on the lubricating oil suction tube bracket (2) with two capscrews (4) and nuts (5).

Tighten the nuts and capscrews (4)(5).

Torque Value: 45 n•m [33 ft-lb]

Tighten the capscrews supporting the main bracket (2). The capscrews **must** be at the tube end of the slot.

Torque Value: 45 n•m [33 ft-lb]

Tighten the top support strap, if applicable (12).

Torque Value: 45 n•m [33 ft-lb]